Topic 5 – Light and the electromagnetic spectrum

Students should:		Maths skills
5.1P	Explain, with the aid of ray diagrams, reflection, refraction and total internal reflection (TIR), including the law of reflection and critical angle	5a, 5b
5.2P	Explain the difference between specular and diffuse reflection	5b
5.3P	Explain how colour of light is related to a differential absorption at surfaces b transmission of light through filters	
5.4P	Relate the power of a lens to its focal length and shape	5b
5.5P	Use ray diagrams to show the similarities and differences in the refraction of light by converging and diverging lenses	5b
5.6P	Explain the effects of different types of lens in producing real and virtual images	5b
5.7	Recall that all electromagnetic waves are transverse, that they travel at the same speed in a vacuum	
5.8	Explain, with examples, that all electromagnetic waves transfer energy from source to observer	
5.9	*Core Practical: Investigate refraction in rectangular glass blocks in terms of the interaction of electromagnetic waves with matter*	
5.10	Recall the main groupings of the continuous electromagnetic spectrum including (in order) radio waves, microwaves, infrared, visible (including the colours of the visible spectrum), ultraviolet, x-rays and gamma rays	
5.11	Describe the electromagnetic spectrum as continuous from radio waves to gamma rays and that the radiations within it can be grouped in order of decreasing wavelength and increasing frequency	1a, 1c 3c
5.12	Recall that our eyes can only detect a limited range of frequencies of electromagnetic radiation	
5.13	**Recall that different substances may absorb, transmit, refract or reflect electromagnetic waves in ways that vary with wavelength**	
5.14	**Explain the effects of differences in the velocities of electromagnetic waves in different substances**	1a, 1c 3c
5.15P	Explain that all bodies emit radiation, that the intensity and wavelength distribution of any emission depends on their temperature	5c
5.16P	**Explain that for a body to be at a constant temperature it needs to radiate the same average power that it absorbs**	

Students should:		Maths skills
5.17P	**Explain what happens to a body if the average power it radiates is less or more than the average power that it absorbs**	
5.18P	**Explain how the temperature of the Earth is affected by factors controlling the balance between incoming radiation and radiation emitted**	
5.19P	*Core Practical: Investigate how the nature of a surface affects the amount of thermal energy radiated or absorbed*	1a, 1c, 1d 2a, 2c, 2f 3a, 3c, 3d 4a, 4c
5.20	Recall that the potential danger associated with an electromagnetic wave increases with increasing frequency	
5.21	Describe the harmful effects on people of excessive exposure to electromagnetic radiation, including: a microwaves: internal heating of body cells b infrared: skin burns c ultraviolet: damage to surface cells and eyes, leading to skin cancer and eye conditions d x-rays and gamma rays: mutation or damage to cells in the body	
5.22	Describe some uses of electromagnetic radiation a radio waves: including broadcasting, communications and satellite transmissions b microwaves: including cooking, communications and satellite transmissions c infrared: including cooking, thermal imaging, short range communications, optical fibres, television remote controls and security systems d visible light: including vision, photography and illumination e ultraviolet: including security marking, fluorescent lamps, detecting forged bank notes and disinfecting water f x-rays: including observing the internal structure of objects, airport security scanners and medical x-rays g gamma rays: including sterilising food and medical equipment, and the detection of cancer and its treatment	
5.23	**Recall that radio waves can be produced by, or can themselves induce, oscillations in electrical circuits**	